

Inferring critical transitions from timeseries

Davide Papapicco*

Graham Donovan*

Lauren Smith*

Merryn Tawhai†

*Department of Mathematics, Faculty of Science, University of Auckland

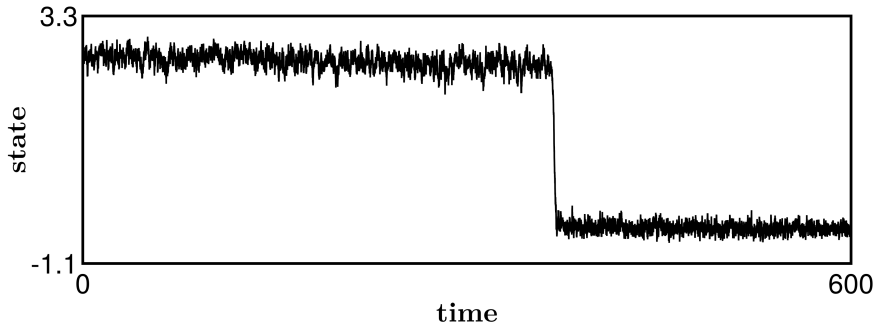
†Auckland Bioengineering Institute



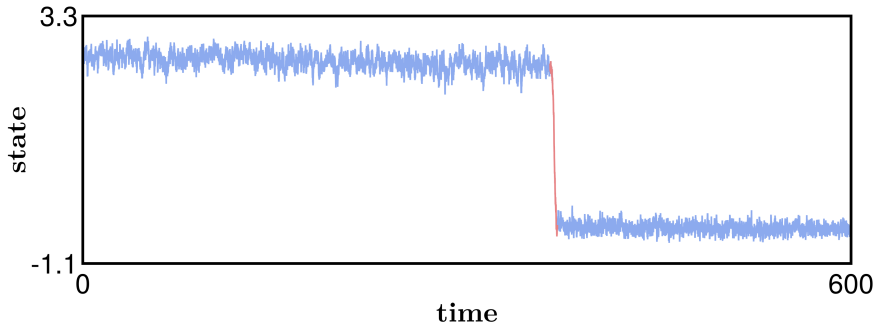
Waipapa
Taumata Rau
**University
of Auckland**



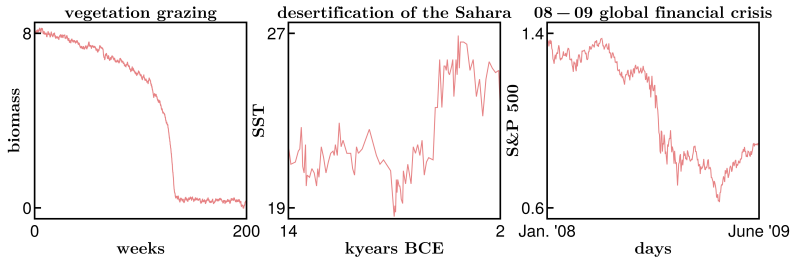
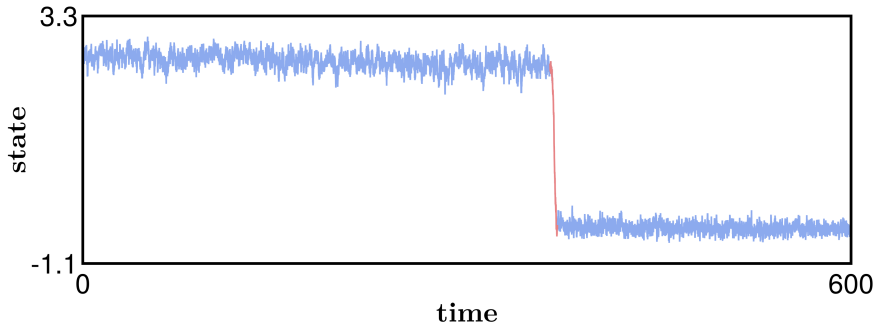
Inferring critical transitions from **timeseries**



Inferring critical transitions from timeseries



Inferring critical transitions from timeseries



Dynamical systems that tip

Consider n –dimensional Itô processes with additive diffusion $\sigma = \sqrt{2D} > 0$ and non-autonomous drift

$$\begin{cases} dX_t = f(X_t, \mu(t))dt + \sqrt{2D}dW_t, \\ \dot{\mu} \in \Lambda_\varepsilon(\mu_-, \mu_+). \end{cases}$$

Dynamical systems that tip

Consider n -dimensional Itô processes with additive diffusion $\sigma = \sqrt{2D} > 0$ and non-autonomous drift

$$\begin{cases} dX_t = f(X_t, \mu(t))dt + \sqrt{2D}dW_t, \\ \dot{\mu} \in \Lambda_\varepsilon(\mu_-, \mu_+). \end{cases}$$

The law of evolution of the parameter is a **parametrised family of shifts** in time connecting the past-limit (i.e. $\mu(t) \rightarrow \mu_-$, $t \rightarrow -\infty$) to the future-limit system (i.e. $\mu(t) \rightarrow \mu_+$, $t \rightarrow +\infty$).

Dynamical systems that tip

Consider n -dimensional Itô processes with additive diffusion $\sigma = \sqrt{2D} > 0$ and non-autonomous drift

$$\begin{cases} dX_t = f(X_t, \mu(t))dt + \sqrt{2D}dW_t, \\ \dot{\mu} \in \Lambda_\varepsilon(\mu_-, \mu_+). \end{cases}$$

The law of evolution of the parameter is a **parametrised family of shifts** in time connecting the past-limit (i.e. $\mu(t) \rightarrow \mu_-$, $t \rightarrow -\infty$) to the future-limit system (i.e. $\mu(t) \rightarrow \mu_+$, $t \rightarrow +\infty$).

For any given μ an **equilibrium** of the system is a state $X^* \in \mathbb{R}^n$ s.t. $f(X^*, \mu) = 0$.

Dynamical systems that tip

Consider n -dimensional Itô processes with additive diffusion $\sigma = \sqrt{2D} > 0$ and non-autonomous drift

$$\begin{cases} dX_t = f(X_t, \mu(t))dt + \sqrt{2D}dW_t, \\ \dot{\mu} \in \Lambda_\varepsilon(\mu_-, \mu_+). \end{cases}$$

The law of evolution of the parameter is a **parametrised family of shifts** in time connecting the past-limit (i.e. $\mu(t) \rightarrow \mu_-$, $t \rightarrow -\infty$) to the future-limit system (i.e. $\mu(t) \rightarrow \mu_+$, $t \rightarrow +\infty$).

For any given μ an **equilibrium** of the system is a state $X^* \in \mathbb{R}^n$ s.t. $f(X^*, \mu) = 0$.

Since $\mu = \mu(t)$ then the equilibrium itself drifts in state space, i.e. $X^* = X_t^*$ is a **quasi-steady equilibrium (QSE)**.

Dynamical systems that tip

Consider n -dimensional Itô processes with additive diffusion $\sigma = \sqrt{2D} > 0$ and non-autonomous drift

$$\begin{cases} dX_t = f(X_t, \mu(t))dt + \sqrt{2D}dW_t, \\ \dot{\mu} \in \Lambda_\varepsilon(\mu_-, \mu_+). \end{cases}$$

The law of evolution of the parameter is a **parametrised family of shifts** in time connecting the past-limit (i.e. $\mu(t) \rightarrow \mu_-$, $t \rightarrow -\infty$) to the future-limit system (i.e. $\mu(t) \rightarrow \mu_+$, $t \rightarrow +\infty$).

For any given μ an **equilibrium** of the system is a state $X^* \in \mathbb{R}^n$ s.t. $f(X^*, \mu) = 0$.

Since $\mu = \mu(t)$ then the equilibrium itself drifts in state space, i.e. $X^* = X_t^*$ is a **quasi-steady equilibrium (QSE)**.

If at any point $\mu(t)$ a QSE is created, destroyed or it changes its stability then a **bifurcation** has occurred.

Tipping mechanisms

We want to differentiate between bifurcation-induced (B-tipping) from rate-induced (R-tipping) critical transitions.

The former type (usually) happens when the parameter shift is too abrupt.

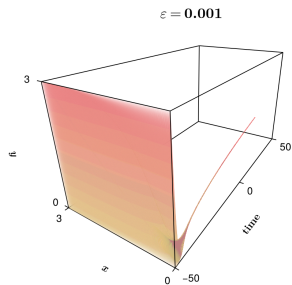
Suppose $\Lambda_\varepsilon(\mu_-, \mu_+) = \frac{\mu_- + \mu_+}{2} + \frac{\mu_+ - \mu_-}{2} \tanh(\varepsilon(t - t_0))$:

Tipping mechanisms

We want to differentiate between bifurcation-induced (B-tipping) from rate-induced (R-tipping) critical transitions.

The former type (usually) happens when the parameter shift is too abrupt.

Suppose $\Lambda_\varepsilon(\mu_-, \mu_+) = \frac{\mu_- + \mu_+}{2} + \frac{\mu_+ - \mu_-}{2} \tanh(\varepsilon(t - t_0))$:

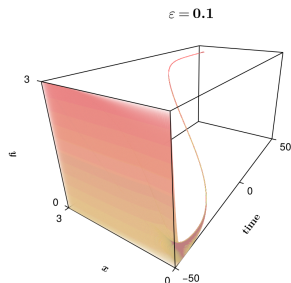
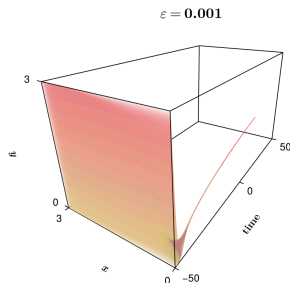


Tipping mechanisms

We want to differentiate between bifurcation-induced (B-tipping) from rate-induced (R-tipping) critical transitions.

The former type (usually) happens when the parameter shift is too abrupt.

Suppose $\Lambda_\varepsilon(\mu_-, \mu_+) = \frac{\mu_- + \mu_+}{2} + \frac{\mu_+ - \mu_-}{2} \tanh(\varepsilon(t - t_0))$:

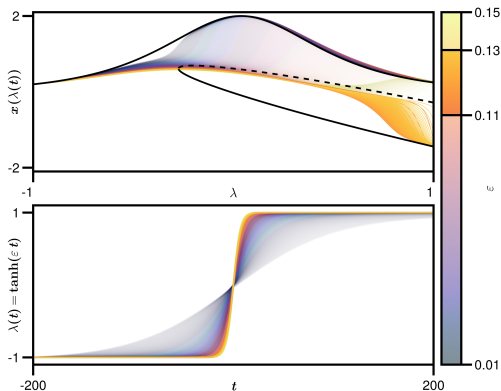


Tipping mechanisms

We want to differentiate between bifurcation-induced (B-tipping) from rate-induced (R-tipping) critical transitions.

The former type (usually) happens when the parameter shift is too abrupt.

Suppose $\Lambda_\varepsilon(\mu_-, \mu_+) = \frac{\mu_- + \mu_+}{2} + \frac{\mu_+ - \mu_-}{2} \tanh(\varepsilon(t - t_0))$:



Inferring critical transitions from timeseries

Consider the following Itô process in saddle-node normal form

$$X_t = X_0 + \int_0^t (-\mu - X_s^2) ds + \int_0^t \sqrt{2D} dW_s.$$

Inferring critical transitions from timeseries

Consider the following Itô process in saddle-node normal form

$$X_t = X_0 + \int_0^t (-\mu - X_s^2) ds + \int_0^t \sqrt{2D} dW_s.$$

Assuming we start the process at fixed $X_0 = \sqrt{-\mu} + \delta$ ($\delta \ll 1$), a solution in a neighbour of $X_t = \sqrt{-\mu}$ is well approximated by the solution of the linear Ornstein-Uhlenbeck process (OUP)

$$X_t = \theta \int_0^t (\sqrt{-\mu} - X_s) ds + \sqrt{2D} \int_0^t dW_s,$$

which admits a stationary probability density with $\theta = -2\sqrt{-\mu}$ and covariance

$$\text{Cov}[X_s, X_t] = \frac{D}{\theta} e^{-\theta(t-s)}, \quad s < t.$$

Inferring critical transitions from timeseries

Consider the following Itô process in saddle-node normal form

$$X_t = X_0 + \int_0^t (-\mu - X_s^2) ds + \int_0^t \sqrt{2D} dW_s.$$

Assuming we start the process at fixed $X_0 = \sqrt{-\mu} + \delta$ ($\delta \ll 1$), a solution in a neighbour of $X_t = \sqrt{-\mu}$ is well approximated by the solution of the linear Ornstein-Uhlenbeck process (OUP)

$$X_t = \theta \int_0^t (\sqrt{-\mu} - X_s) ds + \sqrt{2D} \int_0^t dW_s,$$

which admits a stationary probability density with $\theta = -2\sqrt{-\mu}$ and covariance

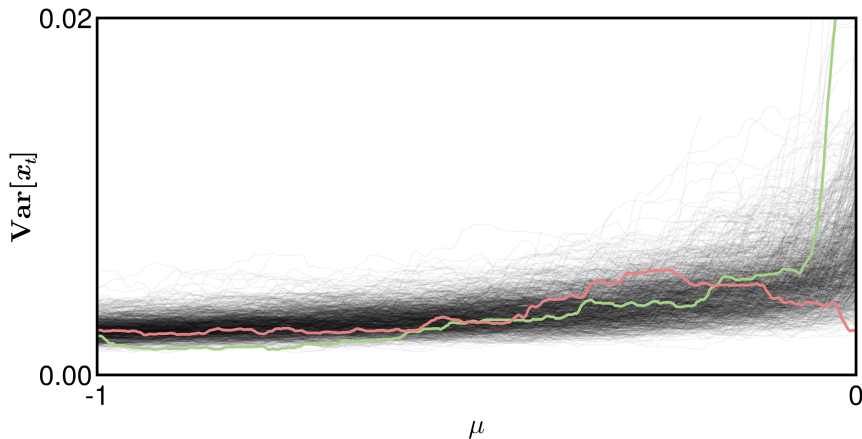
$$\text{Cov}[X_s, X_t] = \frac{D}{\theta} e^{-\theta(t-s)}, \quad s < t.$$

If we now consider instantaneous time increments (i.e. $t \rightarrow s$) then

$$\text{Cov}[X_s, X_t] \rightarrow \text{Var}[X_t] = \frac{D}{\theta} \rightarrow \begin{cases} 0, & \mu \rightarrow -\infty, \\ \infty, & \mu \rightarrow 0. \end{cases}$$

Inferring critical transitions from timeseries

$$f(X_t; \mu) = -\mu - X_t^2 \implies \text{Var}[X_t] = \frac{D}{\theta} \rightarrow \begin{cases} 0, & \mu \rightarrow -\infty, \\ \infty, & \mu \rightarrow 0. \end{cases}$$



Entering large deviation theory

Theorem (Freidlin-Wentzell)

The probability of the sample paths $X^D = \{X_t\}_{t=0}^T$ of $dX_t = f(X_t)dt + \sqrt{2D}dW_t$ to be located in a cylinder of radius $\delta > 0$ around any (absolutely continuous) deterministic path ϕ decays as

$$P[\phi] \asymp \exp\left(-\frac{S[\phi]}{2D}\right) \quad \left(\text{or} \quad \lim_{D \rightarrow 0} \log P\left(\sup_{t \in [0, T]} |X_t - \phi(X_0, t)| < \delta\right) = -S[\phi]\right),$$

where the cost of following such path is given by the rate functional

$$S[\phi] = \int_0^T L(\dot{\phi}, \phi, t) dt = \frac{1}{2} \int_0^T \left(\dot{\phi} - f(\phi)\right)^2 dt,$$

Entering large deviation theory

Theorem (Freidlin-Wentzell)

The probability of the sample paths $X^D = \{X_t\}_{t=0}^T$ of $dX_t = f(X_t)dt + \sqrt{2D}dW_t$ to be located in a cylinder of radius $\delta > 0$ around any (absolutely continuous) deterministic path ϕ decays as

$$P[\phi] \asymp \exp\left(-\frac{S[\phi]}{2D}\right) \quad \left(\text{or} \quad \lim_{D \rightarrow 0} \log P\left(\sup_{t \in [0, T]} |X_t - \phi(X_0, t)| < \delta\right) = -S[\phi]\right),$$

where the cost of following such path is given by the rate functional

$$S[\phi] = \int_0^T L(\dot{\phi}, \phi, t) dt = \frac{1}{2} \int_0^T \left(\dot{\phi} - f(\phi)\right)^2 dt,$$

In the singular limit $\varepsilon \rightarrow 0$ the most likely path $\phi(X_0, t) \in H^1(\mathbb{R}^d, [0, T])$ is the minimiser of $S[\phi]$ and it satisfies the Euler-Lagrange equations (ELEs)

$$\frac{d}{dt} \left(\frac{\partial}{\partial \dot{\phi}} L(\dot{\phi}, \phi, t) \right) - \frac{\partial}{\partial \phi} L(\dot{\phi}, \phi, t) = 0.$$

Treating tipping points as rare events

Corollary

Consider a 1-dimensional, conservative vector field f (i.e. $\exists V$ s.t. $f = -\nabla V$) with a unique, stable fixed point x_0 . Then the probability for hitting a target x scales as $\exp(V(x_0) - V(x))$.

Treating tipping points as rare events

Corollary

Consider a 1-dimensional, conservative vector field f (i.e. $\exists V$ s.t. $f = -\nabla V$) with a unique, stable fixed point x_0 . Then the probability for hitting a target x scales as $\exp(V(x_0) - V(x))$.

Consider a vector field $f(x; \mu)$ with equilibria $x_{\text{stb}, \text{uns}}$.

If the vector field is conservative then its potential $V(x; \mu)$ has a local minima at $a \equiv x_{\text{stb}}$ and a local maxima at $b \equiv x_{\text{uns}}$.

Treating tipping points as rare events

Corollary

Consider a 1-dimensional, conservative vector field f (i.e. $\exists V$ s.t. $f = -\nabla V$) with a unique, stable fixed point x_0 . Then the probability for hitting a target x scales as $\exp(V(x_0) - V(x))$.

Consider a vector field $f(x; \mu)$ with equilibria $x_{\text{stb,uns}}$.

If the vector field is conservative then its potential $V(x; \mu)$ has a local minima at $a \equiv x_{\text{stb}}$ and a local maxima at $b \equiv x_{\text{uns}}$.

Approaching a saddle-node bifurcation ($\mu \rightarrow \mu_{\text{bif}}$) we have $|x_{\text{stb}} - x_{\text{uns}}| = |a - b| \rightarrow 0$, meaning that

$$V(b) - V(a) =: \Delta V \rightarrow 0 \implies \exp(-\Delta V) \rightarrow 1.$$

Treating tipping points as rare events

Corollary

Consider a 1-dimensional, conservative vector field f (i.e. $\exists V$ s.t. $f = -\nabla V$) with a unique, stable fixed point x_0 . Then the probability for hitting a target x scales as $\exp(V(x_0) - V(x))$.

Consider a vector field $f(x; \mu)$ with equilibria $x_{\text{stb,uns}}$.

If the vector field is conservative then its potential $V(x; \mu)$ has a local minima at $a \equiv x_{\text{stb}}$ and a local maxima at $b \equiv x_{\text{uns}}$.

Approaching a saddle-node bifurcation ($\mu \rightarrow \mu_{\text{bif}}$) we have $|x_{\text{stb}} - x_{\text{uns}}| = |a - b| \rightarrow 0$, meaning that

$$V(b) - V(a) =: \Delta V \rightarrow 0 \implies \exp(-\Delta V) \rightarrow 1.$$

We can use the LDP as an EWS!

$$\exp(-\Delta V) = \begin{cases} 0, & \text{if } |\mu - \mu_{\text{bif}}| \rightarrow \infty, \\ 1, & \text{if } |\mu - \mu_{\text{bif}}| \rightarrow 0. \end{cases}$$

Transition probabilities

Given a 1-dimensional conservative system $dX_t = -V'(X_t; \mu)dt + \sqrt{2D}dW_t$ the transition probability $p(X_t, t|X_0, 0)$ is described, over the state space, by the solution of the Fokker-Plank (or forward Kolmogoroff) equation

$$\partial_t p = \mathcal{L}p := (V'p)' + Dp''.$$

Transition probabilities

Given a 1-dimensional conservative system $dX_t = -V'(X_t; \mu)dt + \sqrt{2D}dW_t$ the transition probability $p(X_t, t|X_0, 0)$ is described, over the state space, by the solution of the Fokker-Plank (or forward Kolmogoroff) equation

$$\partial_t p = \mathcal{L}p := (V'p)' + Dp''.$$

Let $\mu \in \mathbb{R}$ be fixed over $\Delta t \gg 0$; if $\frac{\Delta p}{\Delta t} \approx 0$ (i.e. I ignore the transient regime) then the transition probability solves the second-order IVP

$$\begin{cases} Dp'' + V''p + V'p' = 0, \\ p'(\pm\infty) = 0. \end{cases}$$

Transition probabilities

Given a 1-dimensional conservative system $dX_t = -V'(X_t; \mu)dt + \sqrt{2D}dW_t$ the transition probability $p(X_t, t|X_0, 0)$ is described, over the state space, by the solution of the Fokker-Plank (or forward Kolmogoroff) equation

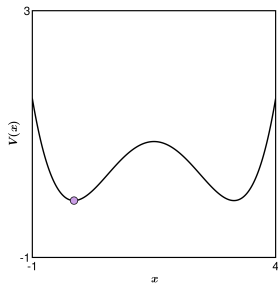
$$\partial_t p = \mathcal{L}p := (V'p)' + Dp''.$$

Let $\mu \in \mathbb{R}$ be fixed over $\Delta t \gg 0$; if $\frac{\Delta p}{\Delta t} \approx 0$ (i.e. I ignore the transient regime) then the transition probability solves the second-order IVP

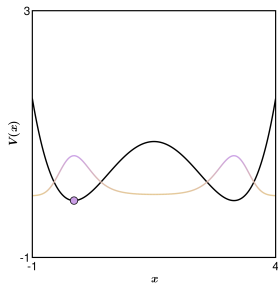
$$\begin{cases} Dp'' + V''p + V'p' = 0, \\ p'(\pm\infty) = 0. \end{cases}$$

I can then solve for $p(x)$ using an integration factor and integration-by-parts to get a **stationary equilibrium density** $p(x) = Z \exp(-V(x)/D)$, with $Z > 0$ being the normalisation constant of $\exp(-V(x)/D)$.

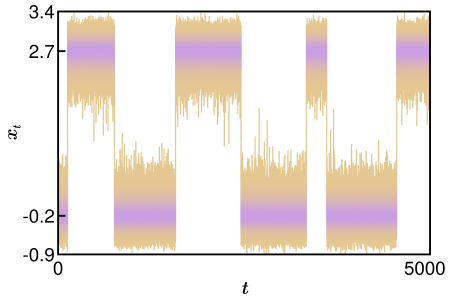
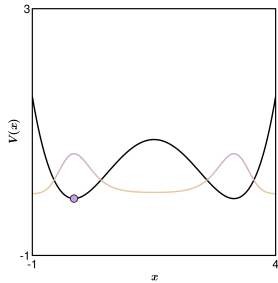
Transition probabilities



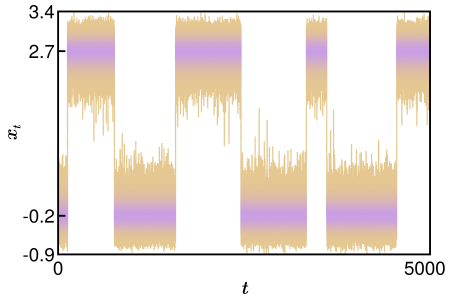
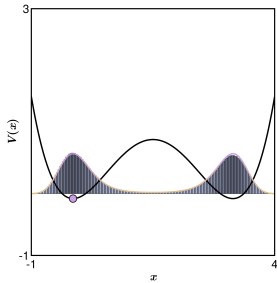
Transition probabilities



Transition probabilities



Transition probabilities



Functional framework

An observed sample path can be thought of as a sequence $\{X_n\}_{n=1}^N \subset I \subseteq \mathbb{R}$ of $N > 0$ i.i.d. random variables sampled from $p(x) \in H := \{f \in L^2(\mathbb{R}) : \int_{\mathbb{R}} f(x)dx = 1\}$.

Functional framework

An observed sample path can be thought of as a sequence $\{X_n\}_{n=1}^N \subset I \subseteq \mathbb{R}$ of $N > 0$ i.i.d. random variables sampled from $p(x) \in H := \{f \in L^2(\mathbb{R}) : \int_{\mathbb{R}} f(x)dx = 1\}$.

Fixing $h > 0$ identifies a unique partition I_h of I into $0 < M < N$ homogeneous (i.e. fixed width), disjoint subintervals

$$I_h = \cup_{k=1}^M B_k, \quad B_k = [\min\{X_t\} + (k-1)h, \min\{X_t\} + kh).$$

Functional framework

An observed sample path can be thought of as a sequence $\{X_n\}_{n=1}^N \subset I \subseteq \mathbb{R}$ of $N > 0$ i.i.d. random variables sampled from $p(x) \in H := \{f \in L^2(\mathbb{R}) : \int_{\mathbb{R}} f(x)dx = 1\}$.

Fixing $h > 0$ identifies a unique partition I_h of I into $0 < M < N$ homogeneous (i.e. fixed width), disjoint subintervals

$$I_h = \cup_{k=1}^M B_k, \quad B_k = [\min\{X_t\} + (k-1)h, \min\{X_t\} + kh).$$

Construct the subspace of H of piecewise constant functions

$$\mathcal{H}_h := \left\{ f \in H : f(x) = \sum_{k=1}^M c_k \mathbf{1}_{B_k}(x) \right\}.$$

Functional framework

An observed sample path can be thought of as a sequence $\{X_n\}_{n=1}^N \subset I \subseteq \mathbb{R}$ of $N > 0$ i.i.d. random variables sampled from $p(x) \in H := \{f \in L^2(\mathbb{R}) : \int_{\mathbb{R}} f(x)dx = 1\}$.

Fixing $h > 0$ identifies a unique partition I_h of I into $0 < M < N$ homogeneous (i.e. fixed width), disjoint subintervals

$$I_h = \cup_{k=1}^M B_k, \quad B_k = [\min\{X_t\} + (k-1)h, \min\{X_t\} + kh).$$

Construct the subspace of H of piecewise constant functions

$$\mathcal{H}_h := \left\{ f \in H : f(x) = \sum_{k=1}^M c_k \mathbf{1}_{B_k}(x) \right\}.$$

Proposition

There exists a unique minimiser of

$$p_h := \operatorname{argmin}_{f \in \mathcal{H}_h} \|f - p\|_H,$$

and it coincides with the bin-averaged true density, i.e. $c_k = \frac{1}{h} \int_{B_k} p(x)dx =: p_k$.

Maximum likelihood estimation

The (empirical) histogram associated to the sample $\{X_n\}_{n=1}^N$

$$\hat{p}_h(x) = \sum_{k=1}^M \hat{p}_k \mathbf{1}_{B_k} = \frac{1}{Nh} \sum_{k=1}^M \mathbf{1}_{B_k}(x) \sum_{n=1}^N \mathbf{1}_{B_k}(X_n),$$

is a random function of $\mathcal{H}_n$ and an unbiased estimator of p_h .

Maximum likelihood estimation

The (empirical) histogram associated to the sample $\{X_n\}_{n=1}^N$

$$\hat{p}_h(x) = \sum_{k=1}^M \hat{p}_k \mathbf{1}_{B_k} = \frac{1}{Nh} \sum_{k=1}^M \mathbf{1}_{B_k}(x) \sum_{n=1}^N \mathbf{1}_{B_k}(X_n),$$

is a random function of $\mathcal{H}_n$ and an unbiased estimator of p_h .

Proposition

The fluctuations between the unbiased density estimator $\hat{p}_h$ and the bin-averaged histogram p_h are asymptotically Gaussian

$$\sqrt{N}(\hat{p}_k - p_k) \xrightarrow{d} \mathcal{N}\left(0, \frac{\pi_k(1 - \pi_k)}{h^2}\right), \quad k = 1, \dots, M,$$

where $\pi_k := \Pr[x \in B_k] = hp_k$ defines the bin probability.

Maximum likelihood estimation

The (empirical) histogram associated to the sample $\{X_n\}_{n=1}^N$

$$\hat{p}_h(x) = \sum_{k=1}^M \hat{p}_k \mathbf{1}_{B_k} = \frac{1}{Nh} \sum_{k=1}^M \mathbf{1}_{B_k}(x) \sum_{n=1}^N \mathbf{1}_{B_k}(X_n),$$

is a random function of $\mathcal{H}_n$ and an unbiased estimator of p_h .

Proposition

The fluctuations between the unbiased density estimator $\hat{p}_h$ and the bin-averaged histogram p_h are asymptotically Gaussian

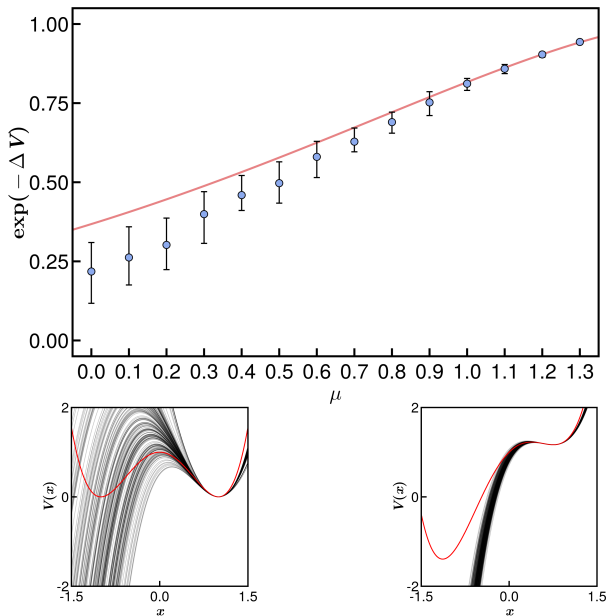
$$\sqrt{N}(\hat{p}_k - p_k) \xrightarrow{d} \mathcal{N}\left(0, \frac{\pi_k(1 - \pi_k)}{h^2}\right), \quad k = 1, \dots, M,$$

where $\pi_k := \Pr[X \in B_k] = hp_k$ defines the bin probability.

Given the (empirical) histogram $\{\hat{x}_k, \hat{p}_k\}$ over the bins' center points $\hat{x}_k$, find

$$\theta^* = (c_j)_{j=0,\dots,3} = \operatorname{argmin}_{\theta \in \mathbb{R}^4} \sqrt{\sum_{k=1}^M (\exp(-U(\hat{x}_k; \theta)) - \hat{p}_k)^2}.$$

Inferring critical transitions from timeseries



Thank you!

-  D. Freedman and P. Diaconis, Probab. Theory Relat. Fields **57** (1981).
-  D. W. Scott, Biometrika **66** (1979).
-  C. Kuehn, Physica D **240** (2011).
-  J. Lohmann and G. Gottwald, Phys. Rev. E **112** (2025).
-  P. Ashwin, et al., Nonlinearity **30** (2017).
-  P. Ritchie and J. Sieber, Chaos **26** (2016).
-  J. Shao, T. Grafke, and R. MacKay, arXiv (2025).

► [Go to the beginning](#)